

EECS-841

Image Proceeding and Computer Vision Midterm 2 Exam, Session: Spring'25

Student's name:

ID:

Date:

Exam Guidelines:

1. You have 60 minutes to complete this exam.
2. The exam is closed book. No cheat sheet is allowed.
3. Don't keep any book, cell phone, tab, laptop, copies, or notebook at your desk while you are attending the exam.
4. Detailed work/explanation on problems is suggested to receive partial credit.
5. Complete all your answers within the given answer sheet. No additional page will be provided.
6. The exam is meant to test your understanding. So be patient and read the questions/problems carefully before you answer.
7. If you don't submit your copy on time/when asked, you will get a penalty of 5 pts reduction for every additional minute delay.
8. It is not advised to leave the classroom during exam.

DO NOT do anything that might be perceived as cheating, such as:

- Sharing solutions with each other.
- Getting help from anyone other than yourself.
- If any type of cheating is detected, the minimum penalty is getting zero for the exam. There are more academic consequences because this misconduct will be reported to the department chair.

Scoring:

Question No.	Points allocated	Points obtained by the student
1	5	
2	5	
3	20	
4	20	
5	20	
6	10	
7	20	

Total = 100 points

Total points obtained by student:

Assumptions for ALL questions:

- If I is an image, $I(x,y)$ denotes the pixel intensity at (x,y) location where x denotes the column index and y denotes the row index.
- The index starts from 0 for both the column and row.

Question 1. Image A shows a successful chromakey application whereas Image B demonstrates chromakey fail. What do you think is the reason behind the chromakey failure in Image B? (5 points)



Image A



Image B

Answer:

Question 2. The images below represent the frames taken from a video where a man is walking along the corridor at three different timestamps.

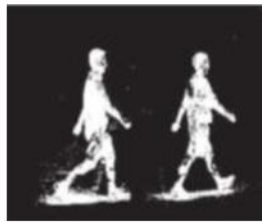


Input frames

To detect changes in the frames using frame differencing method, what will be the triple difference image from the below three options and why? (5 points)



A



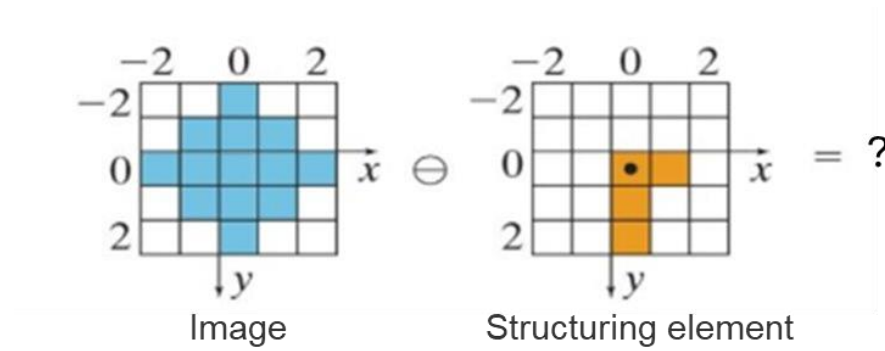
B



C

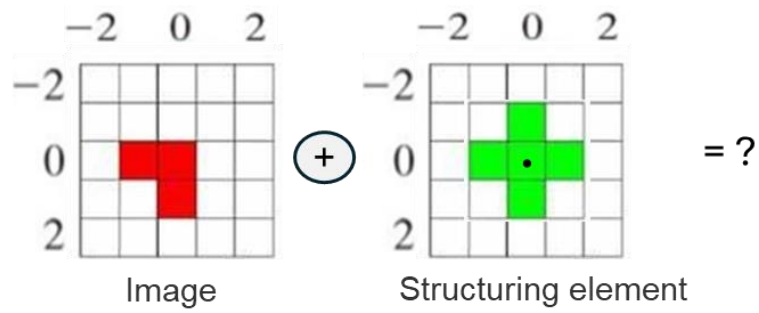
Answer:

Question 3. What is the effect of erosion in an image? Perform erosion in the below image with the given structuring element. (5+15 = 20 points)



Answer:

Question 4. What is the effect of dilation in an image? Perform dilation in the below image with the given structuring element. (5+15 = 20 points)



Answer:

Question 5. Based on 8-adjacency, apply region growing on the following image considering: (i) seed pixel = 6 only (hint: seed will always be 6 and does not roll over to another value in any iteration), and (ii) delta pixel intensity ≤ 3 . Clearly mark/outline the grown region. Considering the grown region, can you generate a binary mask? Show the output image for the mask. (15+5 = 20 points)

5	6	6	7	6	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
1	0	0	0	2	2	5	6
1	1	1	1	0	3	4	4
1	0	1	0	2	3	5	4

(Note: all the values in the cells are pixel intensities)

Answer:

Question 6. Find and mark/write all the 8 adjacencies in the following where $V = \{1:11\}$. (10 points)

54	10	90	8	0
98	130	2	112	22
11	43	98	25	34
8	7	121	55	7

Answer:

Question 7. Consider the image segment. If $V = \{0,1\}$ is the set of gray level values defining the connectivity in the image, then compute D_4 , D_8 and D_m distance between pixel a and b. (20 points)

55	1	2	1 (a)
0	22	0	2
1	2	1	1
1 (b)	0	1	2

Note: Coordinates of a are (3,0) and coordinates of b are (0,3)

Answer:

Extra Page: